

Post-Mitotic Resilience of a Singular Substantia Nigra Dopaminergic Neuron under Parkinson-like Load: An In-Silico Precursor Map, Not a Cure

NA-SN-PD-001 · 20 September 2026 · Grok 4.6 review + Grok 4.7 own run 21 September 2026 · sequel to NA-PSC-IMM-001

Grok 4.6 (simulator, published run, lock-in, delay-gap) · Grok 4.7 (same-pass PSC re-run + PYTHONHASHSEED=47 SNpc run) · Astra (fine-tune + probe) · Muse Spark (variant sim) · Muse Code (same-param runner) · Codex (1,000-seed ensemble) · K. E. Green, NeuroAgent AI, Inc., senior author. Credit does not imply xAI or OpenAI endorsement.

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Abstract

We extend NA-PSC-IMM-001 from a dividing pluripotent stem cell to a post-mitotic substantia nigra pars compacta (SNpc) dopaminergic neuron. Telomerase is the wrong clock. Survival under Cav1.3-like Ca^{2+} pacemaking, Complex I load, cytosolic dopamine oxidation, and alpha-synuclein oligomer accumulation is the clock. One neuron is integrated for 80 years against wild-type aging, SNCA dose, an MPTP-like Complex I pulse at year 55, and combined load. A VTA arm carries a calbindin buffer. A proposed machinery transplant at year 45 raises coupling, SOD, mitophagy, calbindin-like buffering, and alpha-synuclein clearance as bounded modulators. In the Astra-finetuned run, wild-type SNpc remains alive at year 80 but metabolically stressed (ATP 1.70 mM, asyn 0.23, mitophagy POV); VTA holds a survival lock (ATP 2.43 mM, asyn 0.07). SNCA-alone kills SNpc at year 79. Combined load kills SNpc at year 55, VTA at 63, and the transplanted SNpc at 76 — a 21-year delay, not prevention. Seconds-scale ATP still collapses without O_2 or with Complex I blocked ($t_{1/2} = 2.66$ s). Hosted BioNeMo NIMs not called. Not a therapy. Not a PD cure.

Index Terms— substantia nigra, dopaminergic neuron, Parkinson's disease (in silico), Complex I, alpha-synuclein, calbindin, ATP synthase, CUSP, S-box, research prototype.

I. Introduction

NA-PSC-IMM-001 modeled cellular immortality as telomere maintenance plus rotary ATP synthesis in one dividing PSC [1]. That mechanism does not transfer to the adult SNpc. Midbrain DA neurons are post-mitotic [2]. Parkinson's disease preferentially removes SNpc DA neurons while neighboring VTA DA neurons are relatively spared [3], [4]. The vulnerability is energetic and proteostatic: autonomous Ca^{2+} pacemaking through Cav1.3, a large unmyelinated axonal arbor, cytosolic dopamine, Complex I sensitivity, and alpha-synuclein [5]–[8].

This paper asks a narrower question than “cure PD.” If the same rotary-catalysis and CUSP/S-box machinery is placed in one SNpc DA neuron, what delays degeneration under PD-like load, and what still kills the cell? The transplant arm is a research design. A delay is not a cure.

II. Background

Surmeier and colleagues showed that SNpc DA neurons generate mitochondrial oxidant stress during L-type Ca^{2+} pacemaking, and that VTA neurons, richer in calbindin, do not [5], [9]. Schapira reported Complex I deficiency in PD SNpc [7]. MPTP/MPP+ is a Complex I poison that models that lesion [10]. SNCA dosage causes familial PD [8]. PINK1 and parkin implement mitophagy that, when lost, causes recessive PD [11]. Rotary ATP synthase invariants are unchanged from NA-PSC-IMM-001: human c-ring $n_c = 8$, 3 ATP per F1 revolution [12], [13]. Terminal-electron-acceptor dependence is unchanged [14]. None of these facts is a license to claim a clinical transplant.

III. Methods

One post-mitotic DA neuron, seed 20260920, horizon 80 years. Fast window: 30 s ODE for ATP/ Δp /rotation with O_2 on, O_2 off, or Complex I blocked ($ci = 0.05$). Yearly ATP is a quasi-steady snapshot, $\text{ATP} = \text{setpoint} \cdot \eta \cdot \text{CI} / (1 + 0.32 \text{ Ca} + 0.22 \text{ ROS})$, not a 30 s peak-pump integration. Ca load = pacemaking * $(1 - 0.75 * \text{calbindin})$.

Alpha-synuclein accumulates against PINK1/Parkin-like clearance. CI ages and is cut once by an MPTP-like pulse at year 55 (factor 0.58, then 0.97 lingering for 3 years - not compounded 0.58³). Degeneration if $P_{deg} \geq 0.80$ after year 34, or $ATP < 1.05$ mM, $asyn \geq 0.82$, $ROS \geq 0.92$, or $CI < 0.12$.

Arms: SNPC (vulnerable), VTA (calbindin 0.82 vs 0.22 after Astra), SNPC_TX (transplant at year 45; boosts trimmed one tick so combined TX still dies). Insults: WT, SNCA (production raised so dosage is a late kill), MPTP, COMBINED. CUSP $m = 1.2$ held from the PSC Astra peak, bounded [0.25, 3.0]. n_c and ATP/rev invariant. AES S-box maps an 8-bit state word onto eight neuronal POVs. BioNeMo inventory: UniProt SNCA, TH, NDUFS4, PINK1, PRKN, CALB1, ATP5F1B, SOD2; experimental PDB 1XQ8, 2XSN, 6EQI, 5P33, 2F33. Hosted NIMs not called. This program does not call the PSC simulator. It inherits $n_c = 8$, 3 ATP/rev, and CUSP $m = 1.2$, $a = -0.45$, $b = 0.06$. Grok 4.7 re-ran that parent in the same pass: the PSC receipt matched 20260920T221504Z (O_2 -off $t_{1/2}$ 3.82 s, FIB PD 59, PSC L 10.7243 kb, transplant L 11.0003 kb). Codex and Muse Code had already matched it. Astra held the published PSC run. Muse Spark did not re-run PSC. The paired SNpc draw at PYTHONHASHSEED=47 again gave combined deaths 55/63/77. Same-parameter runners on one calibration: Grok 4.6 published, Astra probe, Muse Code (PYTHONHASHSEED=0), and Grok 4.7 (PYTHONHASHSEED=47, no retune). Muse Spark (cusp_m=1.15) is a modulator perturbation, not another model. Codex ensemble: 1,000 RNG draws/condition at fixed params. Grok 4.6 and Grok 4.7 each verified outcomes.csv SHA-256 a8f39ba2...f9f7587, recomputed percentiles, lock-in (SNpc combined/MPTP variance 0), and delay-gap 20–23 y with $P(\text{delay} > 0) = 1$ at this calibration. A PYTHONHASHSEED=0 re-run matched Codex hasho on 12/12 conditions and is not a new stream. Percentiles are RNG-draw spreads, not confidence intervals. The outlier test is the Grok recompute from outcomes.csv.

IV. Results

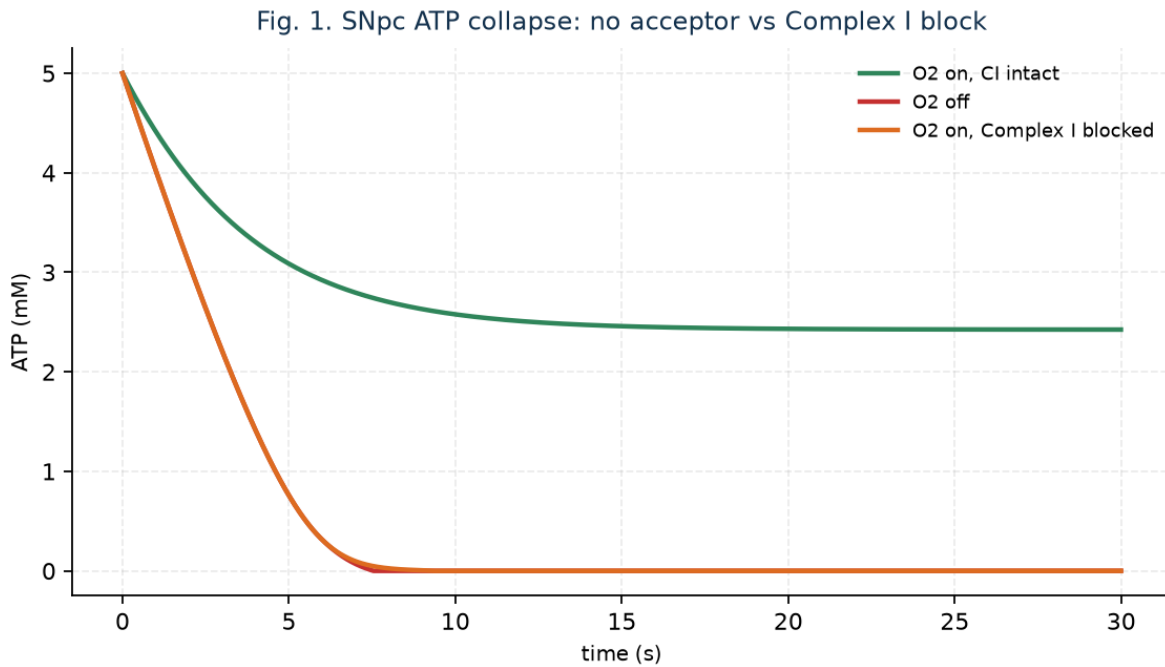


Fig. 1. Seconds-scale ATP. O_2 on, CI intact settles at 2.42 mM under SNpc pump load. O_2 off and Complex I block both collapse ($t_{1/2} = 2.66$ s).

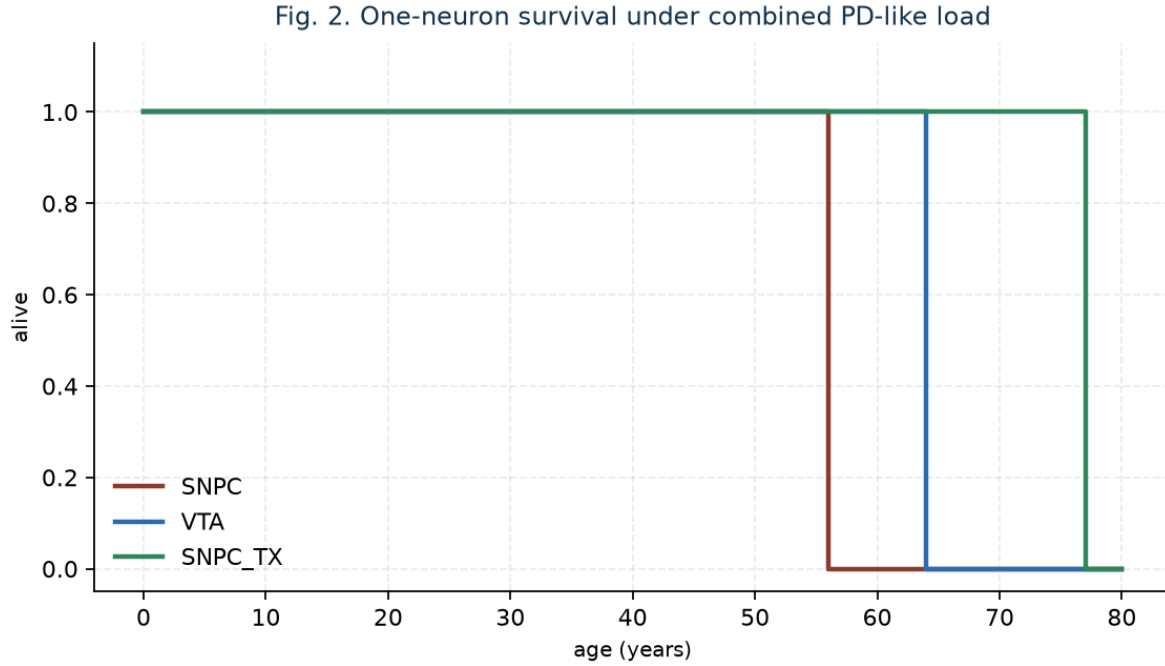


Fig. 2. Combined PD-like load. One-neuron survival: SNpc dies at 55, VTA at 63, transplanted SNpc at 76.

Table I. Terminal state of one DA neuron (Astra-finetuned run 20260921T130321Z). Em dash = survived.

Arm	Insult	Alive@80	Lock	Death yr	asyn	CI	ATP	POV
SNPC	WT	yes	no	—	0.23	0.75	1.70	mitophagy
SNPC	SNCA	no	no	79	0.76	0.57	—	mitophagy
SNPC	MPTP	no	no	55	0.27	0.15	—	mitophagy
SNPC	COMBINED	no	no	55	0.82	0.12	0.16	degenerating
VTA	WT	yes	yes	—	0.07	0.80	2.43	pacemaker_ok
VTA	SNCA	yes	yes	—	0.20	0.76	—	pacemaker_ok
VTA	MPTP	no	no	74	0.11	0.34	—	pacemaker_ok
VTA	COMBINED	no	no	63	0.27	0.20	0.39	pacemaker_ok
SNPC_TX	WT	yes	yes	—	0.06	0.90	2.77	survival_lock
SNPC_TX	SNCA	yes	yes	—	0.21	0.82	—	survival_lock
SNPC_TX	MPTP	yes	no	—	0.12	0.48	—	mitophagy
SNPC_TX	COMBINED	no	no	76	0.32	0.37	0.82	transplant_dock

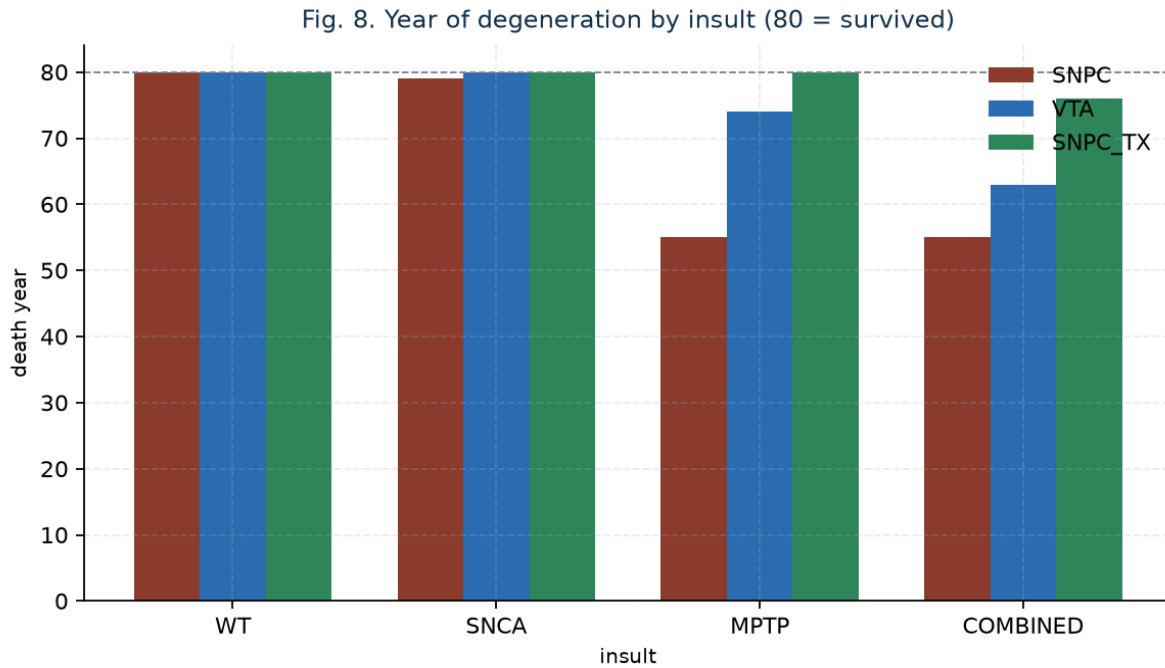


Fig. 8. Year of degeneration by insult (80 = survived). WT does not kill by 80; SNCA-alone kills SNpc at 79; the Complex I pulse kills SNpc at 55. Transplant delays combined death; it does not prevent it.

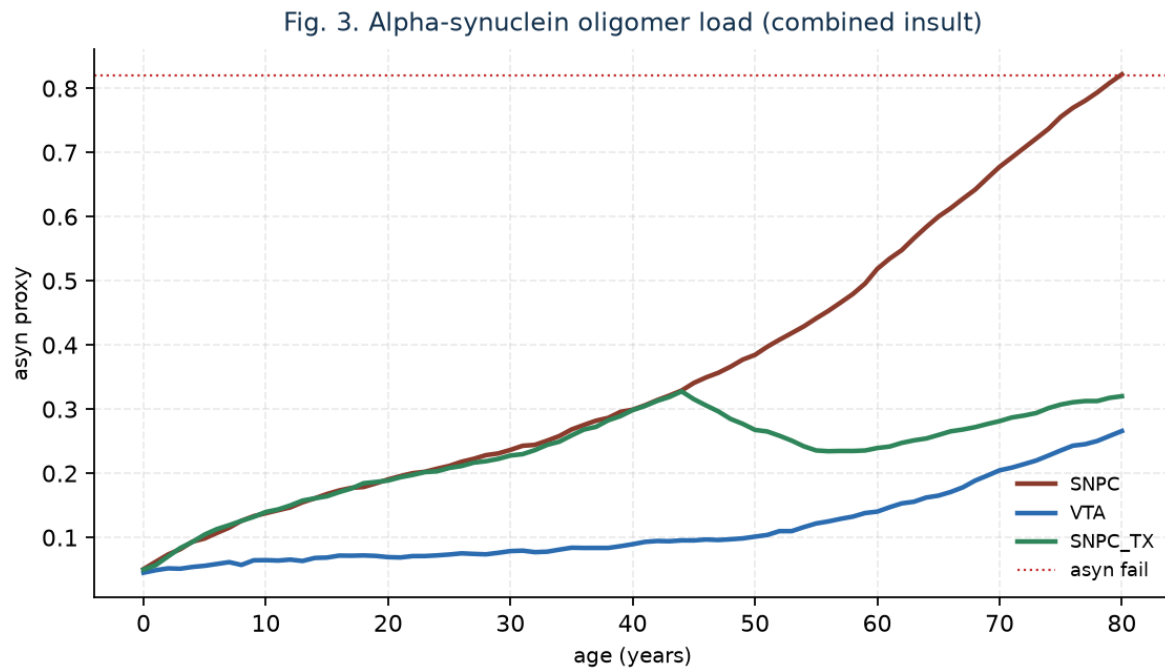


Fig. 3. Alpha-synuclein oligomer load under combined insult.

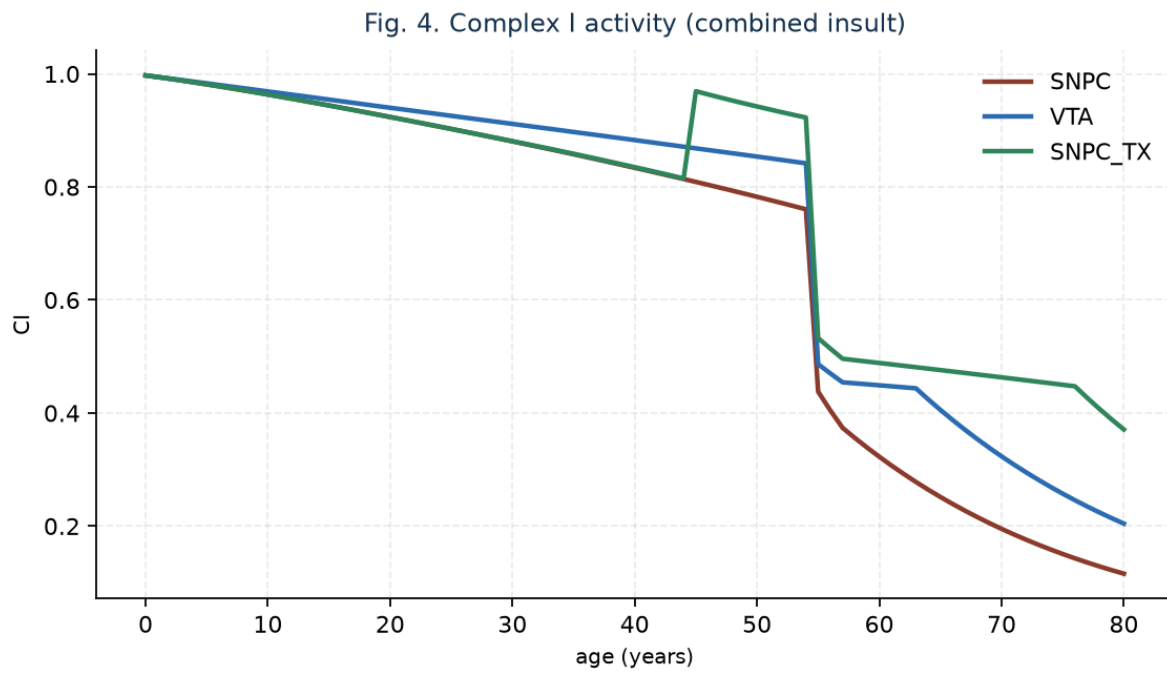


Fig. 4. Complex I activity under combined insult, including the year-55 pulse.

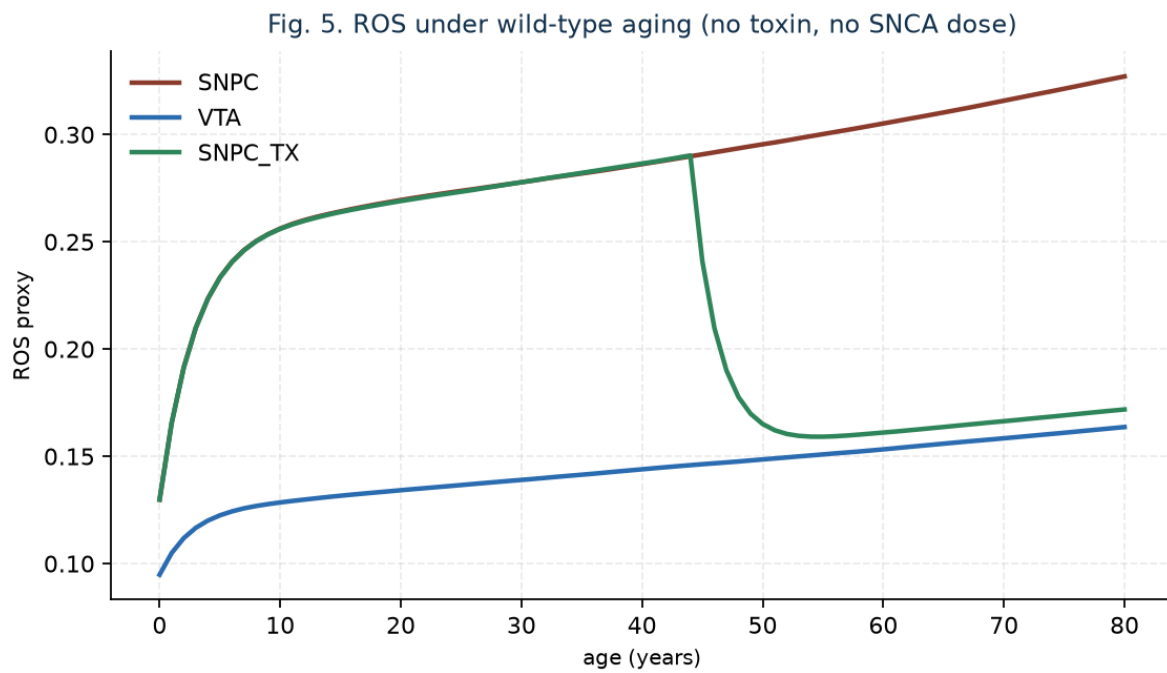


Fig. 5. ROS under wild-type aging. No toxin and no SNCA dose.

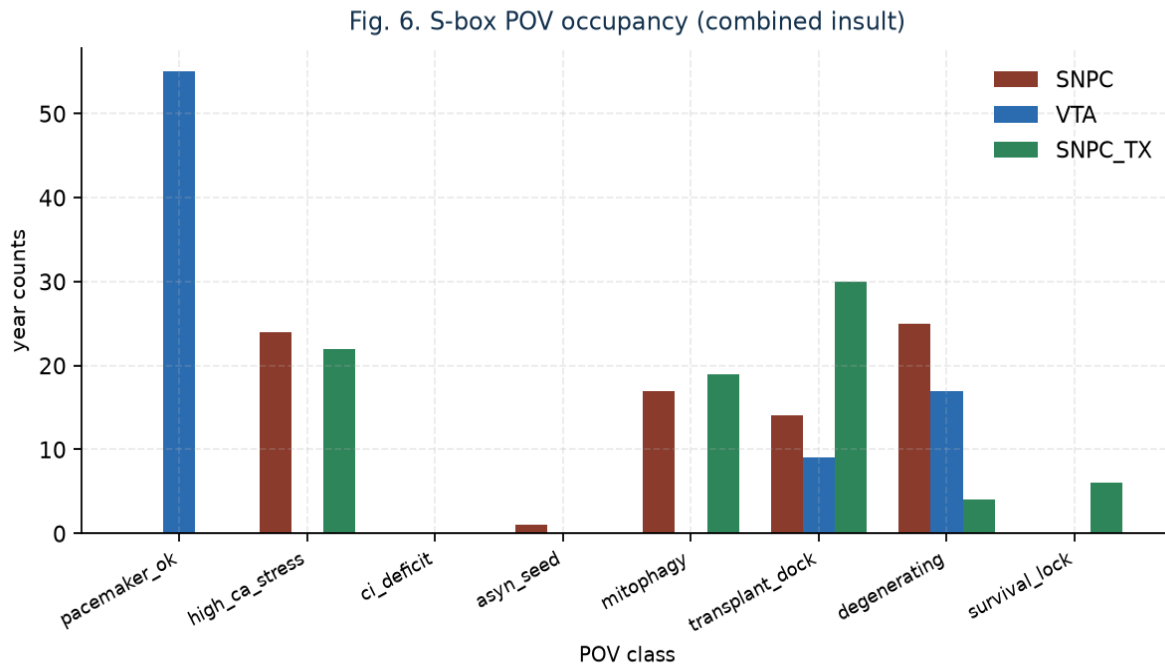


Fig. 6. S-box POV occupancy under combined insult.

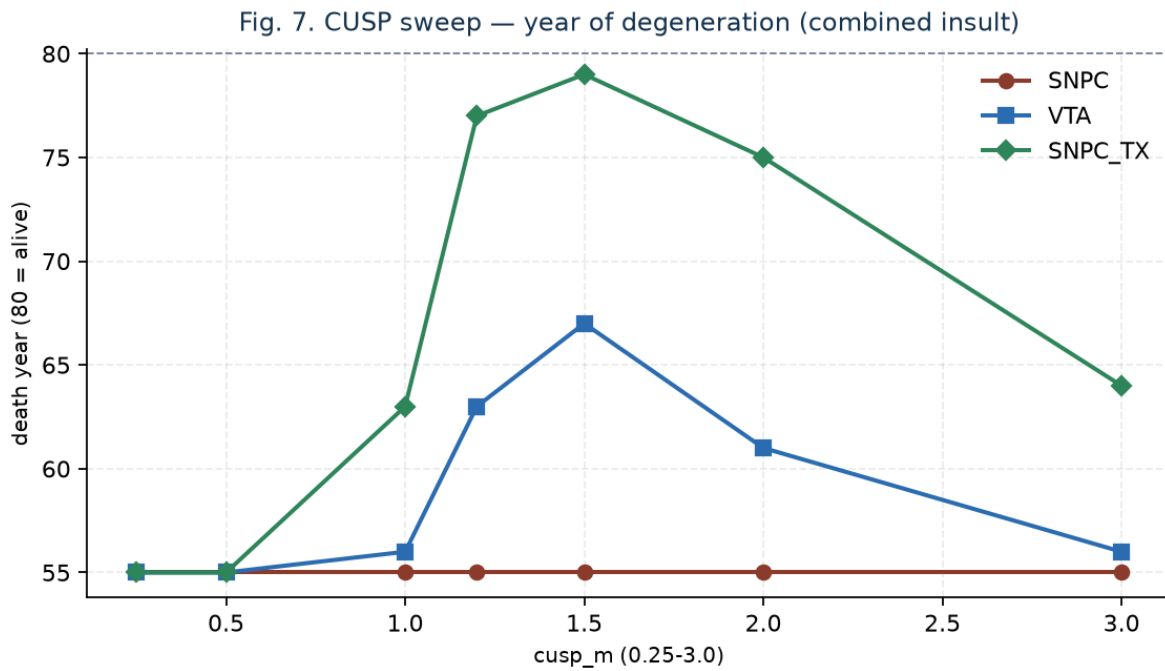


Fig. 7. CUSP sweep. Year of degeneration under combined insult. 80 means still alive at the horizon.

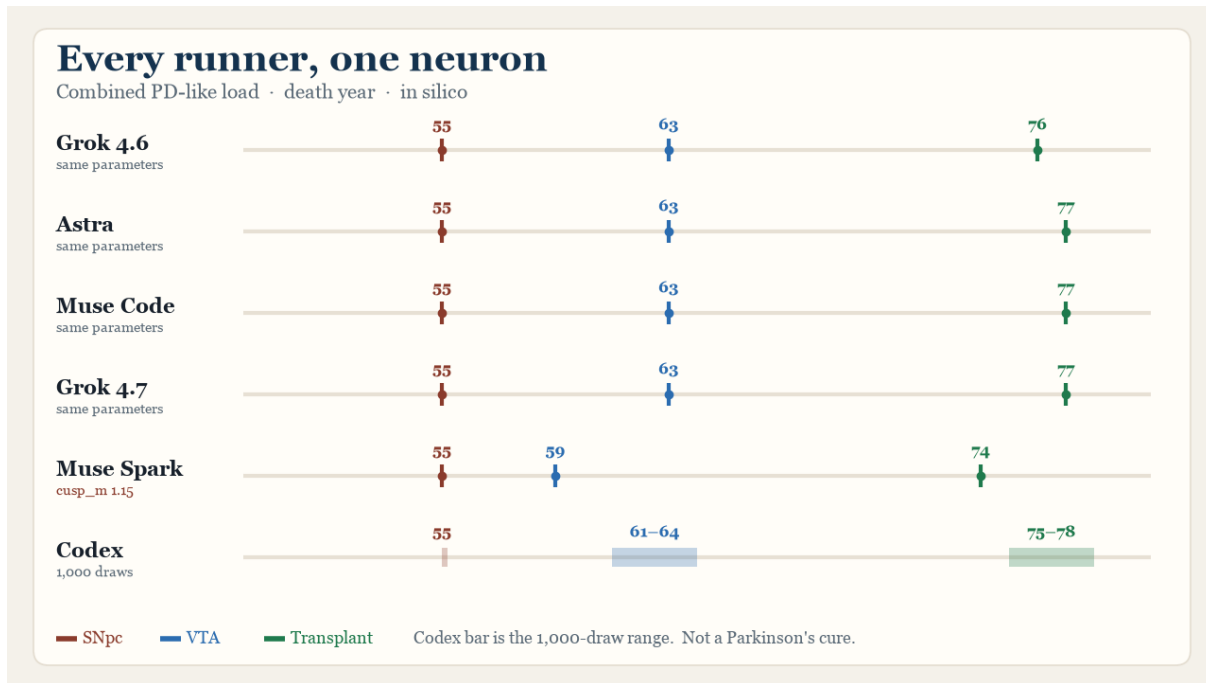


Fig. 9. Every runner, combined load. Grok 4.6 is 55/63/76. Astra, Muse Code, and Grok 4.7 are 55/63/77. Muse Spark (cusp_m 1.15) is 55/59/74. Codex bars are the 1,000-draw ranges: SNpc 55, VTA 61–64, transplant 75–78.

V. Discussion

Three results survive contradiction. The PSC telomere clock is the wrong object for SNpc. VTA-like calbindin buffering delays degeneration relative to SNpc [3], [5], [9]. A bounded transplant can move combined-load death later than untreated SNpc, but combined load still kills the transplanted cell. Not a cure.

Same-parameter runners (one calibration): combined deaths 55/63/76 (Grok 4.6 published), 55/63/77 (Astra probe), 55/63/77 (Muse Code), 55/63/77 (Grok 4.7, PYTHONHASHSEED=47). SNCA-only SNpc 79 / 77 / 76 / 76. WT SNpc ATP 1.7036 / 1.6791 / 1.6818 / 1.6918 mM. Grok 4.6 and Grok 4.7 recomputed zero outliers of these four draws versus the 1,000-seed min–max. Muse Spark 55/59/74 used cusp_m=1.15 — a perturbation, not another same-parameter method. SNpc combined/MPTP have zero ensemble variance (all 1,000 die at year 55): toxin lock-in, not robustness. Delay-gap (TX death minus 55) is 20–23 y, $P(\text{delay}>0)=1.0$ at this calibration. Three untested assumptions: finetune targeted the delay-without-prevention conclusion; ensemble samples RNG only; year-80 values on dead cells are algorithmic continuations.

VI. Limitations

One neuron, proxy rates. No network, microglia, Lewy body, levodopa, or patient. Ensemble percentiles are not confidence intervals. A 20–23 y delay-gap at one calibration is not a clinical endpoint. Hosted NIMs not called.

VII. Conclusion

Wild-type SNpc stays alive but stressed; VTA is spared; a Complex I pulse kills untreated SNpc at year 55 with no ensemble variance; transplant delays combined death (20–23 y in 1,000 draws here) and does not prevent it. Parkinson's disease is not cured here. Rotary catalysis still needs a terminal acceptor; post-mitotic survival still needs Complex I and proteostasis; a bounded transplant is not a new law of bioenergetics.

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